

Flavonoid Glycosides Isolated from Unique Legume Plant Extracts as Novel Inhibitors of Xanthine Oxidase



Legumes and the polyphenolic compounds present in them have gained a lot of interest due to their beneficial health implications. Dietary polyphenolic compounds, especially flavonoids, exert antioxidant properties and are potent inhibitors of xanthine oxidase (XO) activity. XO is the main contributor of free radicals during exercise but it is also involved in pathogenesis of several diseases such as vascular disorders, cancer and gout. In order to discover new natural, dietary XO inhibitors, some polyphenolic fractions and pure compounds isolated from two legume plant extracts were tested for their effects on XO activity. The fractions isolated from both *Vicia faba* and *Lotus edulis* plant extracts were potent inhibitors of XO with IC₅₀ values range from 40–135 µg/mL and 55–260 µg/mL, respectively. All the pure polyphenolic compounds inhibited XO and their K_i values ranged from 13–767 µM. Ten of the compounds followed the non competitive inhibitory model whereas one of them was a competitive inhibitor. These findings indicate that flavonoid isolates from legume plant extracts are novel, natural XO inhibitors. Their mode of action is under investigation in order to examine their potential in drug design for diseases related to overwhelming XO action.